

Hariesh Kumar S K

Senior Generative AI Engineer | AI Solutions Architect | Responsible AI Engineer

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Professional Summary

Senior Generative AI Engineer with 3.5+ years of experience designing and deploying enterprise-grade AI systems from multi-repository agentic knowledge bots and autonomous telecom workflows to governed LLM pipelines on GCP and Azure. Specialized in LLM engineering, RAG systems, multi-agent orchestration, LangGraph workflows, and Responsible AI governance. At Prodapt, I lead telecom-focused GenAI initiatives, building scalable agentic systems that automate ticket processes and reduce operational cost. Passionate about creating safe, auditable, and business-impactful AI with strong guardrail and RAI compliance.

Work Experience

Senior Software Developer — GenAI Engineer | **Prodapt** April 2025 – Present

Leading GenAI initiatives in the telecom domain — delivering high-impact agentic systems and cloud-based AI workflows on GCP using LlamaIndex, LangGraph, RAG, and Vertex AI.

- Architected a multi-repository agentic telecom knowledge bot using LlamaIndex + Vertex AI Search, Git, SharePoint (MS Graph API), data and PDFs into a single governed AI interface — delivering.
- Built a WTN-driven multi-agent troubleshooting system (GCP-native): Master Agent orchestrating Document, Metrics, and Recommendation agents over Pub/Sub — combining Vertex AI Vector Search hybrid retrieval and Gemini LLM for ranked next-best-action suggestions.
- Developed an automatic multi-agent workflow using LangGraph to analyze telecom tickets, classify fault types, automate decisioning, and reduce technician dispatch costs — with human-in-the-loop approval gates and full decision traceability logging in BigQuery.
- Implemented automated email notifications and workflow completion updates via Microsoft Graph API across enterprise pipelines.
- Standardized Responsible AI guardrails, safety scoring, hallucination-reduction, and RAI compliance across all internal GenAI workstreams — including input validation, output filtering, confidence-scored fallbacks, and tool-calling permission boundaries.
- Conducted leadership demo of the agentic knowledge bot, receiving strong organizational visibility and directly influencing AI adoption strategy.
- Received the **Rising Rookie Award** (2025) for outstanding contributions to the multi-repository agentic knowledge project.

Programmer Analyst | **Cognizant** September 2022 – April 2025

Started as a tester, gaining strong QA and workflow optimization skills before transitioning into AI development.

- Developed a RAG-based chatbot for the Jira team using LangChain and FAISS, enabling natural language querying of project data and sprint documentation — awarded the **Spotlight Award**.
- Led organization-wide GenAI knowledge-transfer sessions, upskilling cross-functional teams on LLM fundamentals, RAG architecture, and Responsible AI practices.
- Represented the team in multiple hackathons, delivering high-impact AI proof-of-concept systems that advanced enterprise AI adoption conversations at leadership level.
- Implemented AI workflow improvement initiatives, reducing manual reporting overhead through automated data extraction and summarization pipelines.

Technical Skills

LLM Models: OpenAI, Claude, Gemini, Meta LLaMA, Mistral, Groq, Azure OpenAI, Vertex AI

AI Frameworks & Orchestration: LangChain, LangGraph, LlamaIndex, CrewAI, Hugging Face, Vertex AI SDK, Google ADK, n8n, Flowise

AI Engineering: RAG Pipelines, AI Agents, Tool Calling, MCP Integration, Prompt Engineering, Few-shot & Chain-of-thought, Memory Management, Context & Token Optimization, Fine-Tuning, Multimodal AI, AI Evaluation, Hallucination Reduction, Autonomous Workflows

Cloud & Infrastructure: GCP (Vertex AI, BigQuery, Cloud Run, Pub/Sub, Firestore, GCS, Cloud Scheduler), AWS, Azure, Docker, Docker Compose

Vector Stores & Databases: Vertex AI Vector Search, FAISS, ChromaDB, Pinecone, Elasticsearch (BM25), PostgreSQL, Redis, Firestore, BigQuery

Vision & Multimodal: Stable Diffusion, ControlNet, ComfyUI, LoRA, OpenCV, Google Vision API, OCR, Whisper

Developer Tools & AI Workflow: Claude Code, Claude Coworker, Cursor, GitHub / Git, FastAPI, Streamlit, Flask, Gradio, REST APIs, WebSocket

Responsible AI & Governance: Guardrail Design, RAI Compliance, Safety Scoring, Hallucination Reduction, AI Observability, Human-in-the-loop Systems, AI Governance Frameworks, Ragas Evaluation

Programming Languages: Python, JavaScript, SQL, Bash, Java

Projects

WTN-Driven Telecom AI — Multi-Agent Troubleshooting System

Stack: Master Agent, LangGraph, Vertex AI Vector Search, Elasticsearch, Gemini, textembedding-gecko, BigQuery, Firestore, Pub/Sub, Cloud Run, FastAPI, Streamlit, Ragas

- Architected a GCP-native Master Agent orchestrating Document, Metrics, and Recommendation agents over a Pub/Sub async event bus — enabling fault-tolerant multi-agent coordination at production scale.
- Built hybrid retrieval combining Vertex AI Vector Search (semantic) improving precision for telecom-specific WTN troubleshooting queries.
- Implemented Ragas-based evaluation pipeline measuring faithfulness, answer relevancy, and context precision across troubleshooting sessions.

Multi-Repository Agentic Telecom Knowledge Bot

Stack: LlamaIndex, Vertex AI Search, LangGraph, MS Graph API, SharePoint, Jira, ServiceNow, Watson, GCP Cloud Run, RAG, Responsible AI, ChromaDB

- Architected a multi-index RAG system with a custom LlamaIndex router directing queries across 6 enterprise systems based on query intent
- Integrated MS Graph API for live SharePoint and Teams data; built custom connectors for Watson, ServiceNow, and Git with schema-aware parsing and Responsible AI guardrails on every response.

YukiOS POC — Local-First Autonomous Multi-Agent AI Operating System

Status: In Progress · Personal project

Stack: FastAPI, Python, SQLAlchemy, PostgreSQL, Redis, Groq API, Claude API, Discord.py, Telegram, Docker Compose, WebSocket, Multi-agent, Pydantic, asyncpg

- Architected a local-first multi-agent OS with named specialist agents (Hermes, Paperclip, Ruflo, Security Agent) communicating over a Redis pub/sub event bus — decoupled from cloud APIs.
- Built a real-time WebSocket dashboard, 8-table normalized PostgreSQL schema, Claude Code CLI orchestration layer with human approval gates, and a 16-check pre-flight test harness.

RAG-Based Chatbots for Documents & Websites

Stack: LangChain, FAISS, ChromaDB, OpenAI, Streamlit, Hybrid retrieval

- Built a reusable RAG framework with hybrid dense + BM25 retrieval, source citation enforcement, and configurable chunking — achieving over 85% context retrieval accuracy.

AI Teaching Assistant for Assignment Evaluation

Stack: OCR, OpenCV, Vision LLMs, OpenAI API, LangChain, ChromaDB, Gradio

- Reduced grading time by **60%** through rubric-aligned structured feedback pipelines with longitudinal performance tracking and adaptive difficulty scoring.

Front-End AI UI Generator

Stack: Python, Groq API, Flask, Streamlit, Advanced Prompt Engineering

- Reduced UI development time by **60%** using a multi-stage intent extraction pipeline with sandboxed live preview and iterative conversational refinement.

AI Product Photography System

Stack: Stable Diffusion, ControlNet, LoRA, ComfyUI, OpenCV, Google Vision API

- Built an end-to-end vision AI pipeline generating studio-quality product photography at scale using ControlNet depth conditioning, Stable Diffusion inpainting, and LoRA style fine-tuning.

AI Storytelling Assistant & Multipurpose LLM Chatbot

Stack: LangChain, Stable Diffusion, ControlNet, LoRA, Groq API, Streamlit, Gradio, Hugging Face Spaces

- Multimodal storytelling system with narrative memory management and character-consistent Stable Diffusion visuals. Model-agnostic chatbot platform supporting OpenAI, Gemini, and Claude with pluggable tool registry.

Achievements & Certifications

- **Rising Rookie Award** — Prodapt (2025) — Recognized for outstanding contributions to the multi-repository agentic telecom knowledge project.
- **Spotlight Award** — Cognizant (2024) — Recognized for building a RAG-based chatbot that significantly enhanced knowledge retrieval efficiency for client demos and internal applications.
- **Applied GenAI Certification** — 100xEngineers — Completed hands-on training in LLM-based AI development and Generative AI solutions, covering prompt engineering, fine-tuning, and RAG-based systems.

Education

- **B.Tech** | Amrita Vishwa Vidyapeetham | First Class with Distinction | CGPA: 8.3/10 | 2018 – 2022
- **12th** | Holy Trinity School | 82.3% | 2016 – 2018